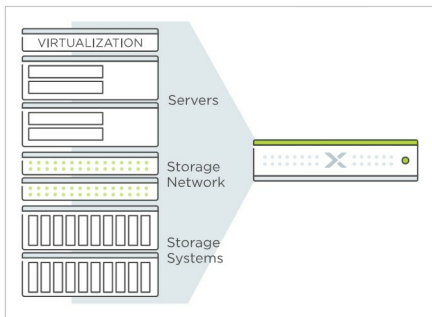




Hyper-convergence combines computing, storage, networking and virtualisation into one easy-to-handle software-defined system (Courtesy: Nutanix)

Finally, even though HCI is a software-defined architecture that brings the convenience of cloud to your enterprise infrastructure, it is not a term that can be used synonymously with cloud computing. The two are quite different. Cloud is a service—it provides infrastructure, platform and software as a service to those who need it. It could be private, public or hybrid, and herein starts the confusion, as many people mistakenly think that hyper-convergence and private cloud are one and the same. Hyper-convergence is merely a technology that enables easy deployment of infrastructure-as-a-service in your private cloud by addressing many operational issues and improving cost-efficiency. It is an enabler, which prepares your organisation for the cloud era, and not the cloud itself!

The bring-your-own-hardware kind

According to a recent Gartner report, software-only “bring-your-own-hardware” hyper-converged systems have become significant and are increasingly competing with hyper-converged hardware appliances.

Gartner research director Jon MacArthur explained in a web report that “Most of the core technology is

starting to shift to software. If you buy a vSAN ReadyNode from Lenovo or Cisco, or HPE, or your pick of server platform, they are all pretty much the same. Customers are evaluating vSAN and ReadyNode, not so much the hardware. Dell EMC VxRail is a popular deployment model for vSAN, but so is putting VMware on your own choice of hyper-converged hardware.”

Special needs like those of the military and the sophistication level of users are some of the factors that influence the decision in favour of implementing HCI as a software stack rather than a pre-packaged solution.

Major players

Gartner predicts that the market for hyper-converged integrated systems (HCIS) will reach nearly \$5 billion (24 per cent of the overall market for integrated systems) by 2019 as the technology becomes mainstream.

Nutanix, a pioneer in this space, offers a range of hardware systems with different numbers of nodes, capacities of memory and storage, different processors, etc. These can run VMware vSphere or Nutanix’s own hypervisor system called Acropolis. The company excels at supporting large clusters of HCI

deployments—scaling up to 100 node clusters—which are easy to use and manage.

Nutanix tempts customers with the idea of a full-stack infrastructure and platform services through “One OS, One Click.” Prism is a comprehensive management solution, while Acropolis is a turnkey platform that combines server, storage, virtualisation and networking resources. Calm is for application automation and lifecycle management in Nutanix and public clouds, while Xpress is designed for smaller IT organisations. Xi Cloud Services help extend your data centre to an integrated public cloud environment. Nutanix also has a community edition that lets you evaluate the Nutanix Enterprise Cloud Platform at no cost!

Companies like Cisco and HPE are strengthening their foothold by acquiring start-ups in the space. Cisco bought Springpath, whose technology powers Hyperflex—Cisco’s fully-integrated hyper-convergence infrastructure system. HPE acquired Simplivity, one of the major competitors of Nutanix. HPE Simplivity products come with a strong, custom-designed platform called OmniStack, which includes a host of features like multi-site data management, global de-duplication, backup, snapshots, clones, multi-site data replication and disaster recovery and WAN optimisation.

Pivot3 also finds a place amongst the most popular HCI players. Last year, the company launched Acuity, which it claims to be the first priority-aware, performance-architected HCI solution with policy-based management. Acuity’s advanced quality of service (QoS) makes it possible to simply and predictably run multiple, mixed applications on a single platform.

Dell EMC offers VxRail and XC series of HCI solutions based on its PowerEdge servers powered by Intel processors. Last year, it released hyper-converged appliances that include Intel’s 14-nanometre Xeon SP processors, along with VMware’s vSAN and EMC’s ScaleIO. Dell EMC offers Nvidia GPU accelerators in both VxRail and XC solutions.